

Fastest jet plane

- **What?** Lockheed SR-71 Blackbird
- **Who?** Clarence "Kelly" Johnson
- **Where and when?** US, 1964

Military jets can fly three times as fast as sound. The Lockheed SR-71 could cross the Atlantic Ocean in less than two hours and, on July 28, 1976, it reached 2,193.2 mph (3,529.6 km/h)—the top speed achieved by a jet plane. The SR-71 was a "stealth" plane, carefully shaped and painted with a special "radar-absorbent" black paint to hide it from enemy radar.



Heat-resistant titanium body was lightweight and helped the plane fly at very high speed.



WOW!

Of the world's 20,000 or so commercial jet airliners, more than 11,000 are in the air at any one moment.



Illustration of NASA's X-43A in flight

Fastest powered plane

- **What?** X-43A
- **Who?** NASA
- **Where and when?** US, 2004

Rockets travel at high speeds, but they have to carry a heavy load of liquid oxygen to mix with their fuel. NASA developed superfast scramjets to solve this problem—these scoop oxygen from

the air as they fly. The X-43A scramjet was an experimental, faster-than-sound, unmanned plane, which was first launched in 2001. Only three X-43As were built. The third was tested in 2004, when it screeched through the air at almost 7,000 mph (11,200 km/h).



Solar-powered plane

- **What?** Solar Impulse 1
- **Who?** André Borschberg and Bertrand Piccard
- **Where and when?** Switzerland, 2009

Jet planes cause pollution, so innovators look for alternative ways to power planes, including solar power. Swiss pilots André Borschberg and Bertrand Piccard made a breakthrough with *Solar Impulse 1*,

which combined rechargeable batteries with solar cells on its wings. The next model, *Solar Impulse 2* (above), completed the first ever solar-powered flight around the world in 2016, and also showed that clean technologies can help achieve goals nobody thought possible.

Fastest glider

- **What?** DARPA Falcon HTV-2
- **Who?** DARPA and US Air Force
- **Where and when?** US, 2011

Imagine traveling around the world in less than two hours. That's how fast the DARPA Falcon Hypersonic Test Vehicle 2 (HTV-2) rocket-launched glider can fly. It is carried high up in the sky by a rocket before gliding back to Earth. It is an experimental craft, but during a test flight in 2011, it reached a top speed of 13,000 mph (21,000 km/h) before crashing.



Illustration of HTV-2